LISTS, STRINGS, AND DICTIONARIES Solutions

COMPUTER SCIENCE MENTORS 61A

February 24, 2026 – February 27, 2026

Created by Anaisha Das and Tavisha Patel. Worksheet feedback: forms.gle/q67BZrHHfoKbF6V28

1 Lists

1. What would Python display? Draw box-and-pointer diagrams for the following:

```
>>> a = [1, 2, 3]
>>> a
```

```
[1, 2, 3]
```

```
>>> a[2]
```

```
3
```

```
>>> a[:2]
```

```
[1, 2]
```

```
>>> len(a)
```

```
3
```

```
>>> c = [x * 2 for x in a]
>>> c
```

```
[2, 4, 6]
```

```
>>> sum(c)
>>> max(c)
>>> all(c)
>>> all([0, 1, 2, 3])
```

```
12
6
True
False
```

```
>>> b = [1, 2, 3, a, 4]
>>> b
```

```
[1, 2, 3, [1, 2, 3], 4]
```

```
>>> b[3][2]
```

```
3
```

2. Write a list comprehension that accomplishes each of the following tasks.

   (a) Square all the elements of a given list, `lst`. First, write a for loop that accomplishes this task using the **range** function. Then, write a for loop using the **in** keyword. Finally, convert it into a list comprehension.

   ```
   [x ** 2 for x in lst]
   ```

   (b) Compute the sum of all even numbers in the list `lst`. Is there a function you can apply to a list to get the sum of all elements?

   ```
   sum([x for x in lst if x % 2 == 0])
   ```

   (c) Return a list of lists such that `a = [[0], [0, 1], [0, 1, 2], [0, 1, 2, 3], [0, 1, 2, 3, 4]]`.

   ```
   a = [[x for x in range(y)] for y in range(1, 6)]
   ```

   (d) Return the same list as above, except now excluding every instance of the number 2: `b = [[0], [0, 1], [0, 1], [0, 1, 3], [0, 1, 3, 4]])`.

   ```
   b = [[x for x in range(y) if x != 2] for y in range(1, 6)]
   ```

3. Complete the definition for `all_true`, which takes in a list `lst` and returns `True` if there are no False-y values in the list and `False` otherwise. Make sure that your implementation is recursive. What is the name of the built-in Python function that does this?

   ```
   def all_true(lst):
       '''
       >>> all_true([True, 1, 'True'])
       True
       >>> all_true([1, 0, 1])
       False
       >>> all_true([])
       True
       '''


       if not lst:
           return True
       elif not lst[0]:
           return False
       else:
           return all_true(lst[1:])

       The built-in Python function all does the same thing that all_true
          does!
   ```

2 Strings

1. What would Python display?

```
>>> a = "hello world!"
>>> a[4]
```

```
'o'
```

```
>>> a = "hello world!"
>>> a[6:11]
```

```
'world'
```

```
>>> len(a)
```

```
12
```

```
>>> 'lo' in a
```

```
True
```

3 Dictionaries

1. What would Python display?

```
>>> a = {1: "one", 2: "two", 3: "three"}
>>> a[2]
```

```
"two"
```

```
>>> b = {"five": 5, "six": [1,2,3], "seven": 7}
>>> b["five"]
```

```
5
```

```
>>> b["six"][2]
```

```
3
```

2. Using a dictionary comprehension, complete the function below. It takes a single-argument function `f` and a dictionary `d`, and returns a new dictionary containing only the entries whose keys are even numbers, with `f` applied to their values.

```
def apply_to_even(f, d):
    '''
    >>> apply_to_even(lambda x: x**2, {1:3, 2:5, 4:2, 3:2})
    {2: 25, 4: 4}
    '''
    return {__ : ______ for _____ in _____ if _____________}
```

```
def apply_to_even(f, d):
    return {i: f(d[i]) for i in d if i % 2 == 0}
```